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Searching for a Pulse: Evaluating the Use of Rapid DC Pulses for Diagnosing Battery Health, State-of-Charge, and Safety

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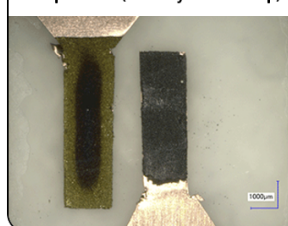
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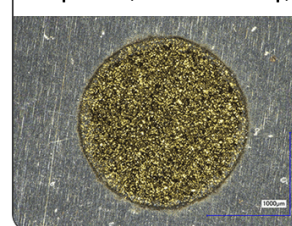
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Sample Test (Side-by-Side Setup)



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Searching for a Pulse: Evaluating the Use of Rapid DC Pulses for Diagnosing Battery Health, State-of-Charge, and Safety

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Rapid electrochemical diagnostics, like DC pulse sequences or electrochemical impedance spectroscopy, are known to be useful for capacity prediction. However, it is unclear how previous results will map to different cell types and use cases and whether rapid diagnostics are useful for remaining useful life prediction or for detecting potential safety issues. To that end, we have collected a data set with ~50,000 DC pulse measurements from four types of commercial lithium-ion batteries to enable training of state-of-charge, health, and safety diagnostic models via machine-learning. We demonstrate that 120-second DC pulse sequences can be used to predict capacity with 2%–9% average error, which can separate high- from low-capacity cells with only a 0.3% false positive rate but is not accurate enough to estimate remaining useful life. We also find that no safety related targets can be accurately predicted, highlighting the critical need for other non-invasive methods to diagnose battery safety.

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The exponential increase of battery systems worldwide requires rapid, scalable, low-cost methods for diagnosing battery state to enable cost-effective repurposing of end-of-first-life electric vehicle (EV) batteries for energy storage applications and to facilitate prognostic health management and preventative maintenance. There are already around 40 million EVs operating globally as of late 2024,¹ or about 3 TWh of battery capacity (assuming 75 kWh average battery pack capacity), with an anticipated 10–20 TWh of capacity from 100–200 million EVs by 2040.^{2,3} Given the huge energy storage capacity of these batteries, repurposing them for stationary storage at the end of their use for transportation presents a huge environmental and economic opportunity. Practically, repurposing these EV cells requires several processing steps, the most onerous and complex of which is to validate their performance and safety prior to their “second life”. Battery repurposing standards, such as UL 1974,⁴ require measuring the open-circuit-voltage (OCV), capacity, DC internal resistance (DCIR), and physical characteristics of every repurposed battery system, which can take anywhere from 6 to 48 hours to complete per battery.⁵

So, given the costs and risks that a battery repurposing facility would incur to perform this entire recertification procedure on end-of-first-life batteries, it is critical that these procedures only be performed on battery systems with a high potential for reuse. To that end, battery repurposing facilities could screen incoming batteries prior to performing recertification tests. A first simple screening step that may be performed even before sending along to a battery repurposing facility is to filter batteries based on the degradation state reported by the battery management system (BMS) or using data from a “battery passport”.^{6,7} Initial screening at the battery repurposing facility should then be done using a non-invasive, rapid, near-zero-net-energy test to prevent wasting of labor and testing resources on unsafe or poor condition batteries. For example, Tao et al.⁸ estimated about a 10% reduction in the total cost to repurpose an end-of-life EV pack and a significant reduction in CO₂ emissions when comparing battery screening using rapid pulses with a battery screening processing using a full charge-discharge cycle. Rapid

electrochemical methods such as AC diagnostics using electrochemical impedance spectroscopy (EIS) or DC diagnostics using a current pulse or sequence of current pulses are well suited to this screening task: measurements only takes seconds or minutes; measurements can be performed using existing pack electrical connections, lowering risk; and the AC or DC signal can be designed to have net zero charge, only consuming energy due to battery pack inefficiency.

Given the need for rapid battery diagnostics, both for battery repurposing but also for online diagnostics, research on the topic of rapid AC and DC diagnostics for batteries has recently surged. The results thus far give some promise to the concept of using rapid electrochemical diagnostics for screening end-of-first-life batteries or for in-operando state estimation. Examples of studies using EIS include capacity prediction^{9–12} and discharge energy prediction from multi-step protocols.¹³ All of these studies report prediction errors between about 1% and 7% (R² between 0.7 and 0.95), and studies using DC pulse sequences seem to report similar accuracy.^{8,14–16} However, drawing a general conclusion as to the effectiveness of these methods is difficult because it is impossible to make like-for-like comparisons: each of these studies used different cells, different characterization protocols, different modeling approaches, varied distributions of battery health, and different approaches to evaluate predictive performance. Furthermore, there is a clear lack of any investigation into diagnosing battery safety and reliability, which is perhaps even more important than diagnosing battery performance from the perspective of a battery repurposing facility. Previous research on safety diagnosis, e.g., fault detection, obviously exists^{17–19} but we do not know of a prior study that experimentally measures safety metrics and attempts to diagnose them without any prior cell history using rapid characterization techniques.

In this work, we have attempted to make a realistic evaluation of the usefulness of rapid electrochemical characterization methods for diagnosing the state of commercial lithium-ion batteries. Using a newly collected data set of about 50,000 independent DC pulse measurements from 259 characterization tests performed at 3 ambient temperatures on 4 different types of commercial lithium-ion batteries of varying manufacturer, chemistry, format, size, and design, we show that 120 second, near-net-zero energy DC pulse

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sequences can be used to estimate state-of-charge (SOC) of NMC-Gr lithium-ion batteries with about 1% mean absolute error (MAE) and predict discharge capacity with 2–9% mean absolute percent error (MAPE). This capacity prediction accuracy is sufficient to sort high (>90%) from low (<80%) discharge capacity cells with less than a 0.3% false positive rate, but is not accurate enough to enable precise remaining-useful-life determination, which requires an accuracy that is better than 1% to be practically useful,⁷ an accuracy that essentially necessitates an actual capacity measurement during the recertification process. A direct comparison using similar evaluation methods to models trained on all open-source EIS-to-capacity data sets we are aware of Refs. (9–12) shows that the accuracy of capacity prediction from AC and DC methods is likely to be equivalent. Safety has been quantified using both electrochemical and physical characterization approaches. In contrast to the positive results for battery state-of-charge and state-of-health prediction, we demonstrate near total failure to predict safety and reliability related targets, like the presence of lithium plating detected by post-charge voltage relaxation or excessive gas generation causing physical deformation of the cell casing. This failure to predict potential safety and reliability risks using rapid DC pulse sequences highlights the critical need for other methods to diagnose safety and long-term reliability of lithium-ion battery systems at scale.

All data, code, and analysis presented in this work is shared open-source, which we hope may be useful to other researchers.

Methods

Electrochemical characterization.—Summarized in Fig. 1, the electrochemical testing protocol was developed to extensively probe state-of-health metrics including capacity and rate testing, pulse response, and application cycle performance. After initial discharge

and a reset cycle, the batteries underwent constant charge - constant voltage (CC-CV) cycles at 3 rates C/10, C/5, and C/3 where the current is held constant at a rate to achieve a full charge for a fully healthy battery over a time period of the denominator, in hours (C/10 is a rate that will yield a full charge in 10 hours). Once the voltage limit is reached, the battery was held at that voltage until the current decayed below C/20. To complete a charge/discharge step, the battery was set to open circuit for 30 min. We then applied a constant power - constant voltage cycle with a rate of P/3. This cycle parallels the CC-CV cycles in every way except the power, not the current, was held constant at a rate based on the rated nominal energy of the battery. Next, we applied a typical hybrid pulse power characterization (HPPC) test to examine pulse response throughout the state-of-charge (SOC) range.²⁰ At each SOC, we performed a series of pulses including 10s–40s–10s discharge-rest-charge pulses at C/10, C/2, 1 C, and 2 C rates. This was followed by 2 sets of predefined pseudo-random pulses (PsRP) where pulses are randomly generated in advance and applied to create a more complex input. After these, the battery is charged or discharged at C/3 for 18 minutes, resulting in an expected 10% SOC change. This is repeated until the voltage limit is hit during the charge/discharge step. After this, we charge the battery to full capacity and apply a US06 drive cycle to simulate a typical vehicle response. This is followed by a charge to 50% SOC and a frequency-containment reserve (FCR) protocol to simulate grid-connected battery coulombic efficiency. We finished the testing with 3 cycles at both C/2 and 1 C. The first was a typical CC-CV as described previously. The next two included dynamic PsRP pulses that maintained the average charge/discharge rate. These 120 s pulse sequences were applied approximately every 10% SOC throughout both charge and discharge cycles.

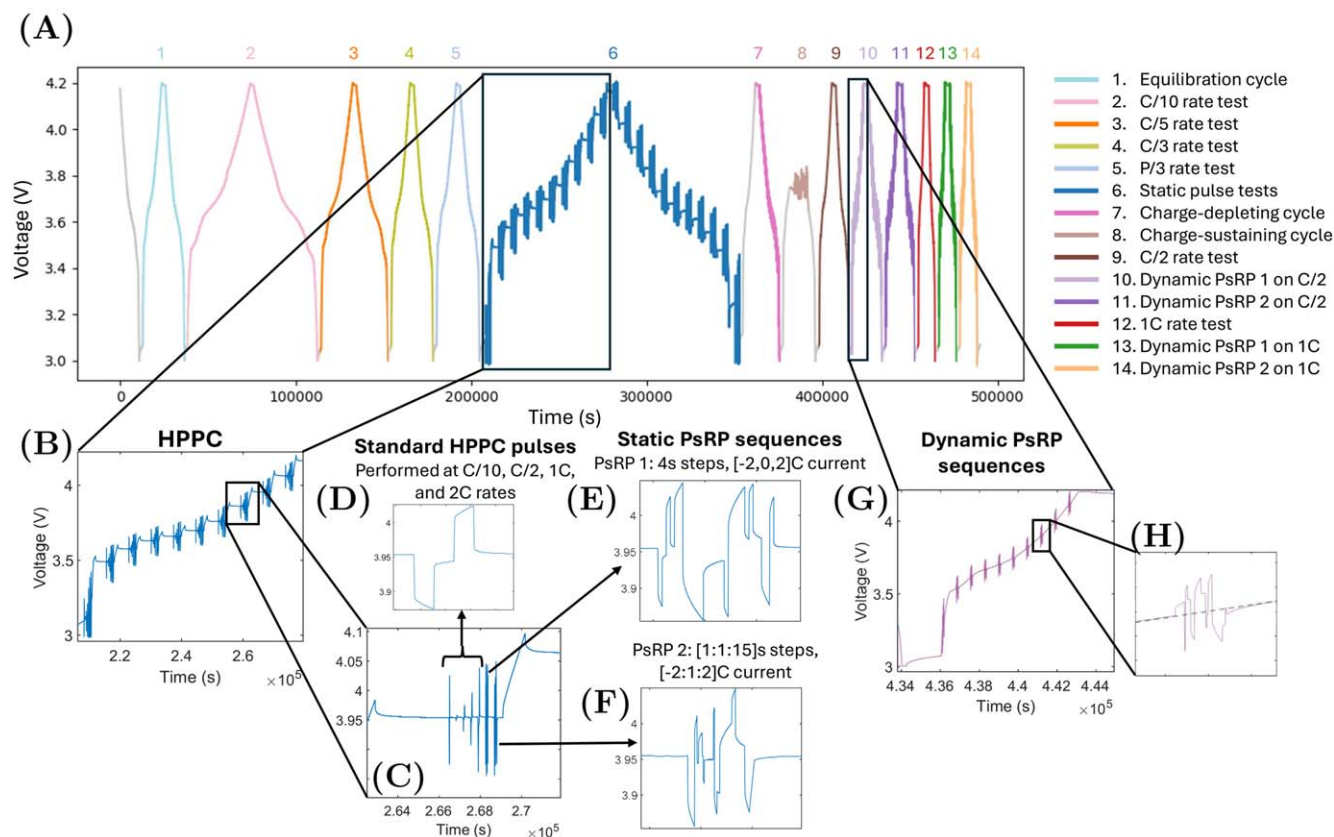


Figure 1. Overview of the full electrochemical characterization test. (A) Voltage timeseries of the entire characterization test, with named segments labeled. (B) Detail of the HPPC Charge portion of the Static pulse tests. (C) Detail of a single set of static pulse tests conducted all at one SOC. (D) Detail of an “HPPC” type pulse, here, a “Rapid” pulse sequence. (E) Detail of the Static PsRP 1 pulse sequence. (F) Detail of the Static PsRP 2 pulse sequence. (G) Detail of the Dynamic PsRP 1 pulses overlaid on a C/2 charge. (H) Detail of a single Dynamic PsRP 1 pulse, with the dotted black line highlighting the steady increase of the average voltage during the pulse sequence.

Pseudo-random pulse (PsRP) sequences were generated by randomly sampling from an input set of pulse times and pulse currents using code documented in the repository to generate pulse sequences 120 seconds long. “PsRP 1” (both Static and Dynamic) pulses used only 4 second long pulse segments and $[-1, 0, 1]$ pulse current rates. “PsRP 2” pulses used one of eight linearly-spaced pulse times between 2 and 15 seconds, and $[-1, -0.5, 0, 0.5, 1]$ pulse current rates. For the pulses with zero DC current bias (i.e., static pulses), a randomly-generated 60 second sequence was simply mirrored to get zero net DC current bias. For pulses with non-zero DC current bias (i.e., the dynamic pulses, which charge or discharge the cell on average), the random sampling of currents was weighted according to the desired DC bias, and random sequences were generated while varying the random generator seed until a pulse sequence with the exact desired DC bias was found.

The supplemental materials of this document contains an Excel spreadsheet example of the A123 electrochemical testing protocol, including the US06 and FCR time-varying power profiles.

This test was applied to each cell at 3 chamber temperatures: 15°, 30°, and 45°C. It was our intention to run these tests from high temperature to low in order to avoid possible Li plating that may occur at low temperatures; however, issues with data collection resulted in an inability to fully maintain this order.

Additional lab aging of Type B and Type C cells.—Cells of types B and C originally had capacities that were clustered in distinct groups corresponding to their pack of origin. To fill in the gaps and create a more continuous capacity distribution, many of these cells were cycled under various temperature and rate conditions. They were then intermittently retested with the entire testing protocol at each temperature. It should be noted here that the computed tomography data for cells of type C were collected at the very end of all testing so it can only be compared to the last testing data collected. As a result, we don’t have physical measurements of many early-life cells of type C.

Extraction of features and targets.—Each measured characterization test, described in Electrochemical characterization section, was deconstructed into a set of inputs (features) and outputs (targets) for the machine learning procedure outlined in Machine learning procedure section. The inputs are the temperature, current, and voltage response during each pulse, of which there are many instances and types during a single characterization test. Common pulse types are stacked together across all measurements and each pulse is relabeled to start from 0s. The outputs are state-of-health metrics which include discharge capacity, US06 drive cycle charge throughput, FCR cycle charge efficiency, and safety; these comprise one unique value per measurement. Discharge capacity is computed using the cumulative maximum absolute capacity during rate capability segments. Charge throughput is calculated as the integral of current over time during the US06 drive cycle segment. Charge efficiency of the FCR segment is calculated as the ratio of the integral of charge current to the integral of discharge current. The calculation of safety during post-charge voltage relaxation is described in detail in Quantification of safety from voltage relaxation section.

Quantification of safety from voltage relaxation.—Physical degradation mechanisms like gas generation and lithium plating pose risks to safety. These mechanisms cause non-ideal cell behavior that is reflected in changes to the shape of the post-charge voltage relaxation curve.²¹ We quantify safety from an electrochemical perspective by estimating how much the post-charge relaxation curves deviate from the “ideal” shape, which is expected to relax according to a simple exponential decay. To do so we fit the first 15 minutes of the post-charge voltage relaxation with a simple exponential, and then extrapolate that curve to the next 15 minutes. The deviation between the extrapolated and observed voltage relaxation curves, computed as the mean squared error, quantifies

how similar the shape of the voltage relaxation curve is to that of an “ideal” battery. Cells that have high deviation from ideal may be considered as “unsafe”. A visual explanation of this safety metric and example values are shown in Supplemental, Figs. 1 and 2.

Physical characterization and X-ray CT scanning.—Physical characterization of type A cells was performed by measuring electrode stack thickness with a sheet micrometer and cell volume using a water displacement column. This process is documented more thoroughly in Gasper et al.²²

Each cell for cell types C and D was individually scanned using the Nikon XT H 225 ST 2x system in Glimpse’s facility (Somerville, MA). This system is equipped with Nikon’s 225 kV rotating anode X-ray source and a flat-panel detector (Varex Imaging, Inc.). The data acquisition was performed using Nikon’s Inspect-X software (version XT 6.12). Scan acquisition settings were set to balance both scan time and image quality; the scan acquisition time was six minutes and two minutes for the prismatic cells and cylindrical cells, respectively. The scans were subsequently processed via Glimpse’s scan processing pipeline, which performs several image enhancement steps including intensity adjustment, cropping, and denoising. These processing steps were consistent across all scans of the same form factor.

We used built-in area and distance measurement tools in the Glimpse Portal to assess cell and electrode volume expansion. To measure volume expansion in the prismatic cell (cell type C), we took a cross-sectional area measurement of the casing at the cell center in both the YZ and XY directions (Supplemental, Figs. 4 and 5); volume growth was calculated by normalizing volumes by the nominal cell volume at beginning of life, which was either measured from a pristine cell or known from manufacturer specifications. We also took cross-sectional area measurements of the jelly roll in both directions as another data point. The jelly roll measurement proved to be a poor predictor of gas generation because of the cell-to-cell heterogeneities in gas occupation: some cells dissipate the gas outside of the jelly roll while some form large internal air pockets. Thus, the full casing cross-sectional area is the better metric for quantifying gas generation. To measure electrode stack expansion, we took the distance from the topmost current collector to the one at the fold in the YZ plane. Averaging this over the 35 dual layers in the roll gives an estimate of the average electrode stack thickness. We performed a sensitivity analysis of this by taking 4 separate measurements of cell cross-sectional area and electrode stack thickness. From this, we found a maximum difference of 2.5% between consecutive area measurements and <1% in stack thickness measurements. Each of these is below the maximum difference between cells, indicating that we can observe significant deviations across our sample set. Finally, we noted the presence/absence of excess electrolyte around the Al current collector overhang. This is easily observed as the X-ray absorption of bulk electrolyte stands out well in contrast to very thin Al foil or simply empty space.

We used similar methods to assess structural evolution in the cylindrical cells (cell type D). We took cross-sectional area measurements for both radial and axial slices through the center of the cell (Supplemental, Figs. 4 and 5). The variation in these measurements across all 30 cells is quite small, emphasizing the difficulty with diagnosing cylindrical cell health and safety metrics with physical measurements. We also measured excess internal volume by tracing the outline of the innermost positive electrode layer in the roll. This measurement is complemented by the stack thickness which we measured from the outermost negative electrode near the end of the active material to the innermost negative electrode that still has active material. These metrics are loosely inversely related which has some physical sensibility: as electrode layers expand with cycling, the excess volume decreases to account for the stack expansion. Moreover, we also observed more qualitative metrics like excess electrolyte presence and electrode roll buckling. The buckling is a strong predictor of poor cell performance while the absence of excess electrolyte in the center pin does not

necessarily indicate a degraded cell. Finally, we measured the negative electrode overhang on both the top and bottom of the roll. No clear trends in battery health were observed from this other than the high variability in the overhang indicating poor control of processing conditions for the manufacturer.

Machine learning procedure.—We used Extreme Gradient Boosting (XGBoost) decision tree models to predict all targets using the XGBoost library in Python.²³ The inputs to the XGBoost models were the voltage (V), current (I), and temperature (T) measurements of the pulses, as well as the computed polarization (P), defined as the voltage at each time step discounted by the initial voltage ($V_t - V_0$). The outputs of the models were the various state-of-health metrics including discharge capacities, Coulombic efficiency and cycle throughput for specific application cycles, state-of-charge, and safety. Safety was predicted as a binary classification, where voltage relaxation fit errors above a certain threshold were considered to be “unsafe” or outliers and those below were “safe” or non-outliers. Discharge capacity categories of “good”/“okay”/“bad” were also predicted as a multiclass classification using a softmax objective loss function. All other targets were predicted using regression with a mean-squared-error objective. For each target, separate XGBoost models were fit on each pulse type to compare their predictive power. We also stratify by cell type before fitting, i.e. fit separate models for different cell types, as the distribution of available capacities vary by orders of magnitude. We experimented with neural network architectures including multi-layer perceptron (MLP), Bi-directional Long Short Term Memory (LSTM), and CNN-LSTM, but found XGBoost to have the highest performance (Supplemental, Fig. 6). Models were compared using the R^2 metric of linearity, mean absolute error (MAE), and mean absolute percentage error (MAPE) for the regression targets and compared using precision, recall, and F1 score when predicting safety and qualitative capacity classifications. For robust estimates of prediction uncertainty, we report the distribution of scores across 50 repeated randomly-sampled 80%/20% train/test splits. Within each train/test split, data is grouped by characterization test for continuity between the input pulses and measured targets.

Results

Overview of the collected data.—We have collected data from four different types of commercial lithium-ion batteries, with varying chemistry, capacity, format, and manufacturer, as described in Table I. We cannot disclose the original manufacturer of every cell type; however, all cells were produced by major manufacturers for either electric vehicle or stationary storage applications. As we are attempting to show relationships between rapid DC pulse response and cell performance metrics like discharge capacity, it was crucial to test a variety of cells at different states of health. This was achieved by using lab aged cells (cell type A²²), field aged cells (cell type D¹²), and field aged cells with additional “pristine” samples and lab aging to fill gaps in the data set (cell types B and C). More details on sourcing of cell types A and D can be found in their respective references (A:²²; D:¹²). Type B cells are Nissan Leaf cells from a 2017 Nissan Leaf with unknown history. Type C cells were extracted from two hybrid vehicle packs that were acquired in 2022 and then disassembled: one from Texas, USA, model year

2017 with 70,000 miles, and one from New York, USA, model year 2014 with 110,000 miles. In the raw data set, these cells are denoted as “C_TX” and “C_NY”, respectively. Additional pristine samples for cell type C were obtained directly from the cell manufacturer and had been kept at low SOC in a -5°C freezer for about 6 years with no measurable changes in properties or performance since acquisition. Every cell performed a 6-day electrochemical characterization test to measure various performance metrics and voltage response to several DC pulse sequences at 15°C , 30°C , and 45°C using an environmental chamber with convective temperature control. Each individual characterization test contains DC pulse sequences performed across the voltage window, giving hundreds of independent capacity measurements and tens of thousands of independent DC pulse sequences (Table I). After collecting data from both cell types B and C, we noticed fairly large gaps in the capacity data, so some cells were cycled and/or calendar aged in lab to result in a more uniformly distributed data set.

Electrochemical characterization.—The electrochemical characterization test measured discharge and charge capacity at various rates, performance during charge-depleting electric vehicle (US06²⁰) and charge-sustaining stationary storage (frequency containment reserve, i.e., FCR²⁴) application cycles, as well as cell response to a variety of rapid DC pulse sequences. An example voltage trace from the electrochemical characterization test is shown in Fig. 1A. Rapid DC pulse sequences were performed both after resting for 30 minutes, as in the standard hybrid pulse power characterization (HPPC) testing procedure and referred to here as “static” pulses (Fig. 1B), as well as “dynamic” pulses that charge or discharge the cell during the pulse sequence (Figs. 1G–1H). The duration and design of static pulses are varied (Fig. 1C), starting with simple 60 s HPPC pulses (10 s discharge, 40 s rest, 10 s charge) at C/10, C/2, 1 C, and 2 C current rates, a “rapid” 16 s simple pulse sequence (4 s discharge, 4 s rest, 4 s charge, 4 s rest) performed at a 2 C current rate (Fig. 1D), and two different pseudo-random pulse (PsRP) sequences that are 120 s long, one randomly sampling step current (Fig. 1E, “Static PsRP 1”) and one randomly sampling both step duration and current (Fig. 1F, “Static PsRP 2”); these two PsRP designs are also replicated for the “dynamic” pulses.

Figure 2 shows the high-level relationships between capacity, DCIR, and temperature for each cell type. DCIR was calculated using Ohm’s law on 2 C HPPC pulses across the SOC range for each characterization test and then interpolated at 50% SOC. Figures 2A–2D show the distributions of relative discharge capacity for each cell type; most samples of cell types A and C measure between 95% and 85% capacity, while cell types B and D are more uniformly distributed with a higher variance, with measurements from near 100% capacity down to below 25% capacity. The overall relationship between DCIR and discharge capacity, in absolute terms, is shown in Fig. 2E. This plot makes clear the different power-to-energy (P/E) ratios of the cell types: cell type C has about half the capacity at the same resistance as type A and B cells, i.e. a higher P/E ratio. All cells have an obvious correlation between capacity and DCIR, giving some confidence that mapping between rapid electrochemical measurements and discharge capacity should be feasible. Note the small number of total measurement counts for cell type B in Fig. 2B and Table I; due to the relatively wide distribution of

Table I. Summary of collected data.

ID	Manufacturer	Format	Chemistry	Capacity	# Batteries	# Tests	# Pulses	Ref.
A	n/a	Pouch	NMC-Gr	64 Ah	16	49	10163	²²
B	Nissan	Module (Pouch)	LMO-Gr	66 Ah	8	28	4074	N/A
C	n/a	Prismatic	NMC-Gr	26 Ah	21	94	19186	N/A
D	A123	Cylindrical	LFP-Gr	2.3 Ah	30	84	14483	¹²
Totals:					79	259	47906	

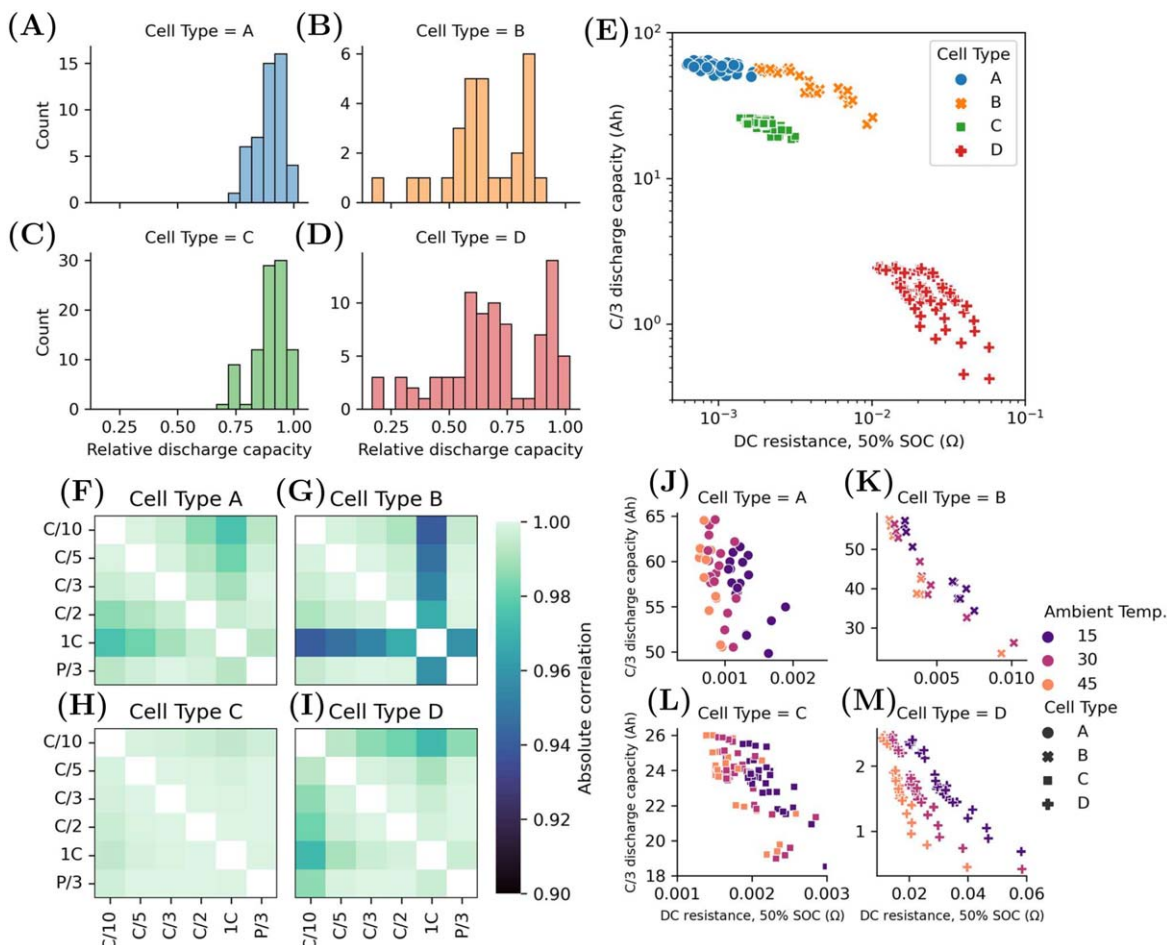


Figure 2. High level summary of the dataset, showing relationships between capacity, DCIR, and temperature across the entire dataset and for each cell type. (A-D) Histograms of discharge capacity for each cell type. (E) Relationship between DC resistance at 50% SOC and C/3 discharge capacity for all cell types. (F-I) Absolute correlation between constant-current discharge capacity at various discharge rates and the constant-power discharge capacity at the P/3 rate. (J-M) Relationship between DC resistance at 50% SOC, C/3 discharge capacity, and temperature.

measured capacities and small number of total measurements for cell type B, we do not think that results shown for type B are likely to be representative of real world diagnostic accuracy or uncertainty.

The effect of P/E ratio and chemistry on the measured discharge capacities at each tested rate are shown in Figs. 2F–2I, using the pairwise absolute correlation to display trends from each cell type. Darker hues indicate lower correlation. Cell types A and B, which have a lower P/E ratio, show the lowest correlation for the high rate measurements, due either to rate limitations or inconsistent cell behavior at high rates. Conversely, cell types C and D show no evidence of being rate-limited at 1 C. Cell type D, the only LFP-Gr cell in this work, shows the greatest variance of performance at the C/10 rate, perhaps due to rate dependent hysteresis^{25,26} or current-driven activation of the iron-phosphate to LFP phase transition.²⁷

The correlations between capacity and DCIR for each cell type independently can be observed in Figs. 2J–2M, as well as the impact of temperature. Cell type A seems to only have a loose correlation between DCIR and capacity, while types B, C, and D all have strong correlations. Cell type D shows the cleanest overall trends. Temperature has a large impact on DCIR but only a small impact on discharge capacity, so any method that uses rapid electrochemical measurements to estimate cell properties needs to be proven to be able to compensate for measurement temperature, or measurements need to be conducted in controlled environments.

Quantification of battery safety from electrochemical response.—

In addition to cell performance, we have also attempted to quantify cell safety from the recorded electrochemical characterization data.

Quantification of safety was inspired by prior work demonstrating the use of 15- to 30 minute voltage relaxation measurements after charging to detect the presence of lithium plating.^{21,28} These works show both from first principles and with experimental evidence that as the extent of lithium plating increases, the voltage relaxation transitions from an “ideal” behavior, which can be modeled almost perfectly using a small number of exponential relaxations,²⁹ to “non-ideal” behavior, that is, a voltage response that cannot be modeled well by a small number of exponential relaxations. In prior works, the identification of “ideal” and “non-ideal” voltage response is done by calculating the time-derivative of voltage and comparing the plotted derivatives. Here, voltage data was recorded on channels with 1 mV resolution, which makes numerical differentiation of the voltage impossible in many cases as some cells only relax by a few mV after charging.

So, to separate ideal battery behavior from non-ideal behavior, i.e., the behavior of a potentially unsafe cell, an exponential relaxation curve is fit to the first 15 minutes of all post-charge relaxations, and then extrapolated to the next 15 minutes of each curve (Supplemental Material, Fig. S1). The mean squared error (MSE) of the extrapolation compared to the observed curve is then taken as a “measurement” of cell ideality. For cell type A, “non-ideal” cells appeared after charging at “high” rates of C/2 and 1 C at 15 °C, suggesting detection of lithium plating; however, this trend was not necessarily followed by other cell types (Supplemental Material, Fig. S2). By plotting the resulting MSEs from each charging rate, we determined that MSE values after charging at rates of C/2 and 1 C were most useful for separating “non-ideal”

cells from “ideal” cells. We then manually identified thresholds to separate “unsafe” from “safe” cells to train classification models to predict this “safety” metric. Thresholds are shown in Supplemental Material Fig. S3, and detailed explicitly in the shared code. Cell type D has no “unsafe” cells, as no outlier MSE values were observed.

Measurement of physical characteristics and X-ray CT scanning.—Many prior works have thoroughly investigated the relationships between cell performance/failure and electrode thickness,^{22,30–32} gas generation,^{22,33–35} excess electrolyte,^{31,36,37} and electrode buckling,^{38,39} highlighting the importance of measuring physical cell properties when evaluating a battery’s future reliability and safety. So, after completion of all electrochemical testing, most cells of types C and D were imaged with X-ray absorption computed tomography (XCT). XCT resolves X-ray absorption differences throughout a cell volume and allows for non-invasive quantification of many physical characteristics or for identification of defects like electrode buckling. Cell types A and B were too large to scan using XCT. A summary of the XCT data for cell types C and D is shown in Fig. 3. For both cell types, physical parameters such as electrode stack thickness and cell volume were measured. We also noted qualitative observations like the presence of excess electrolyte and electrode buckling. Detailed graphical depictions of how electrode stack thickness and cell volume were measured for cell types C and D are in Supplemental Material, Figs. S4 and S5. Physical characteristics of cell type A had been measured previously.²²

Cells of type D showed no measurable volume change, owing to the rigidity of the cylindrical cell geometry with a stabilizing pin in the center. Overall degradation state (capacity loss and resistance growth) did seem related to qualitative observations like the presence or absence of excess electrolyte in the center pin (Figs. 3A, 3B) and the presence or absence of buckling of the electrode roll near the center pin (Figs. 3C, 3D). Electrode thickness is clearly affected by degradation, with obvious electrode thickness growth visible when comparing Figs. 3C and 3D; this layer expansion is likely the root cause of the electrode buckling in Fig. 3D.

Cells of type C showed obvious variation of electrode expansion, total cell volume, and defect presence across the collected XCT measurements. For instance, Fig. 3F shows a cell with significant volume expansion leading to deformation of the cell casing, while Fig. 3G shows a cell with less volume expansion but obvious electrode thickness growth, likely causing the crumpling of the electrode sheets near the middle of the cell leading to an obvious void space filled with excess gas. Excess electrolyte was more difficult to qualify for type C cells than type D; we found that looking for excess electrolyte wetting between Al current collector foils in the bottom corner of the cell seemed to be the most reliable method (Fig. 3E).

Relationships between key quantities.—Figure 4 shows heatmaps of the pairwise absolute correlations between DCIR and all of the key quantities so far discussed. Darker hues indicate smaller correlation. The correlations between DCIR and discharge capacity at 1 C and C/10 agree with the trends observed in Fig. 2: from high correlation between DCIR and C/10 discharge capacity to low, the cells are ranked B-C-D-A, with a maximum correlation of about 90% for cell type B and a minimum correlation of about 40% for cell type A. This implies that, at best, DCIR should be able to predict discharge capacity with high accuracy (90% correlation), and at worst, should always be better than simply guessing (zero correlation). The effect of the discharge rate on the correlation varies across cell types.

Electrode thickness expansion is also relatively well correlated with DCIR (50% for cell types C and D, 40% for cell type A, see Supplemental Fig. S7). This is likely because electrode thickness growth is due primarily to growth of the solid electrolyte interphase (SEI) at the graphite-electrolyte interface, which is the core

degradation mechanism experienced by essentially all Li-ion batteries and directly causes irreversible thickness growth of the graphite electrode and resistance increase.^{30,32} Positive electrode particle cracking may also contribute to electrode stack thickness growth, but has a more complex effect on DCIR throughout the lifetime of a cell.^{22,31,40} Suffice to say, the electrode stack is never irreversibly shrinking as a battery degrades. Even for cylindrical and prismatic cells, this thickness expansion is not usually constrained by the cell can, as getting the jelly roll into the cell can in the first place requires some free space, though some of this free space may be taken up by volume expansion during electrolyte wetting and formation.^{30,41}

The safety and reliability targets like volume growth, post-charge relaxation ideality (“safety”), and excess electrolyte presence are poorly correlated to DCIR. The correlation of cell volume growth and DCIR is mostly dependent on the cell format: cell type A is a pouch cell, explicitly designed with excess pouch material that can expand when gas is generated, so any generated gasses increase cell volume and the correlation to cell performance (DCIR and discharge capacity) is high; cell type C is a prismatic cell, which resists significant expansion but can still deform a little bit, leading to a partial correlation between cell volume growth and cell performance; cell type D is a cylindrical cell, which is much more rigid than the other cell types and can tolerate high internal pressures, leading to almost no correlation between volume change of the cell can and performance. Excess electrolyte seems to have a minor correlation to discharge capacity but is poorly correlated to DCIR, suggesting it may not be detectable using rapid electrochemical measurements. Worst of all, the “safety” target has near-zero correlation with DCIR for cell types A and C, indicating that it may be infeasible to detect cells exhibiting non-ideal behavior using rapid pulses. Cell type B has a decent correlation between DCIR and “safety”, though this may simply be due to a lack of data leading to autocorrelation between “safety” and discharge capacity; only extremely low capacity type B cells (<50% relative discharge capacity) exhibited outlier post-charge voltage relaxation behavior and were considered “unsafe”.

Some limitations of this data set are obvious from Fig. 4. Type B cells are actually small modules, so the physical characteristics could not be measured physically without disassembling the module, which was deemed infeasible for this work, and the module was too large to image via XCT. Type A cells were also too large to image via XCT and so the absence or presence of excess electrolyte could not be determined, though the invasive characterization of some end-of-life type A cells did note a qualitative lack of excess electrolyte after substantial capacity loss.²² As noted in the Methods, no outlier “safety” values were observed for the type D cells, so we decided to ignore the “safety” target for cell type D.

State-of-charge diagnosis using rapid DC pulse sequences.—To begin illustrating the capability of machine learning models using raw I-V-T-t data to perform battery diagnostics, we first demonstrate SOC prediction on all cell types in the data set. This data set provides a large number of feature-target pairs for training SOC diagnostic models, as each cell type has thousands of independent pulse measurements recorded across the entire voltage window at various states of degradation and temperatures (Table I, Fig. 1). The example problem of SOC diagnosis allowed us to test different machine learning architectures, feature engineering methods, and model performance evaluation approaches before training large numbers of models on the many possible feature-target pairs in this data set, without worrying that lack of training data would bias our results toward specific architectures. Best results were achieved using an Extreme Gradient Boosting (XGBoost) machine learning model architecture²³ with the raw current, voltage, and temperature (I-V-T) data as inputs. XGBoost is more often than not the best model architecture for modeling a huge variety of problems using tabular data.⁴² The only additional feature found to reliably improve predicted performance was a “polarization” vector, which is simply

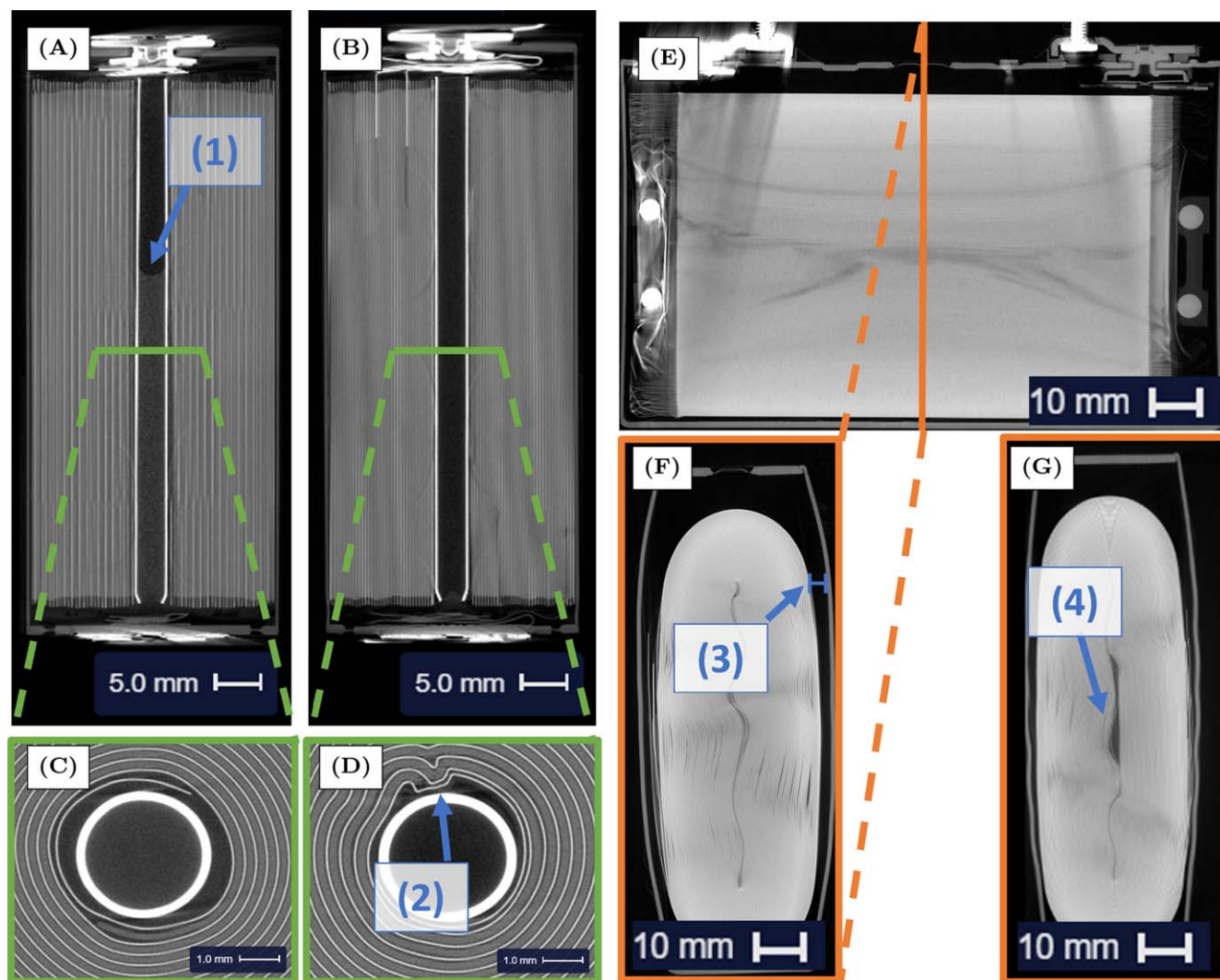


Figure 3. Example computed tomography visualization. (A,C) Cross-sectional views of nearly pristine cylindrical cell (type D) with excess electrolyte (blue arrow, 1) in center pin. The green line in (A) denotes the cross-sectional slice shown in (C). (B,D) Cross-sectional views of highly degraded cylindrical cell with electrode crumpling (blue arrow, 2) and no excess electrolyte in center pin. The green line in (B) denotes the cross-sectional slice shown in (D). Note the stack expansion in the lower-absorbing (darker) negative electrode layers in (D). (E-G) Cross-sectional views of prismatic cell (type C). (F) Cell can deformation due to gas generation (blue arrow, 3). (G) Electrode buckling (blue arrow, 4). Cell Details (Cell ID, relative C/3 discharge capacity at 30 °C): (A,C) D_1, 96% capacity; (B,D) D_19, 17% capacity; (E,F) C_TX_37, 85% capacity; (G) C_NY_28, 75% capacity.

the voltage minus the initial voltage of any given DC pulse measurement. More details on the machine learning approach can be found in the Methods. Performance of models is evaluated on 50 repeated random samples of 80%/20% train/test splits to fairly evaluate the potential accuracy of these models on previously unseen data. Data from each characterization test is grouped during train/test splitting to avoid data leakage.

Figure 5 shows the results of SOC prediction from each cell type. Results using the Static PsRP 1 pulse as input are compared with a baseline model that uses only the open-circuit voltage (OCV), i.e., the initial voltage, and the temperature as inputs; the [OCV, T] model has no information about a cell's dynamic response, so it is basically learning a SOC-OCV-T lookup table. For cell types A and C, SOC is predicted with 1% mean absolute error (MAE) (Fig. 5A) across all random splits, which is about 2–3 x better than the [OCV, T] baseline. The parity plots for these cells show almost perfect correlation between the actual and predicted SOC, with just slightly higher average error in the middle of the SOC range (Figs. 5B, 5D). The accuracy of these SOC estimates is comparable to reported accuracy of real-world battery management systems for NMC-Gr cells.⁴³ Much more challenging, though, is the SOC estimation for

type D cells, where the flat OCV-SOC curve of the LFP-Gr cell makes SOC estimation extremely challenging. The [OCV, T] baseline model results in an average MAE of 8%, 3–8x worse than for the other cell types. Even adding information about cell kinetic behavior by using the Static PsRP 1 pulse as input only decreases error to 5%, which is too large to be useful. The parity plot for cell type D (Fig. 5E) shows maximum errors of almost 30% in middling SOC regions, and even 25% maximum error at true SOC's close to 0% and 100%, showing the challenge of LFP-Gr SOC estimation across the entire voltage window.

Diagnostic accuracy depends on the target.—The performance of the XGBoost models on a variety of different targets, using the Static PsRP 1 data as input, is shown in Fig. 6. Performance is reported using MAPE and R^2 (Figs. 6A–6C). As a reader, consider both MAPE and R^2 when evaluating a model's performance, as they measure different aspects of the relationship between a data distribution and a predicted distribution: MAPE summarizes the proportional magnitude of model errors, while R^2 summarizes the proportion of observed variance in the target that is explained by the model. A high-performing model should have good MAPE and R^2

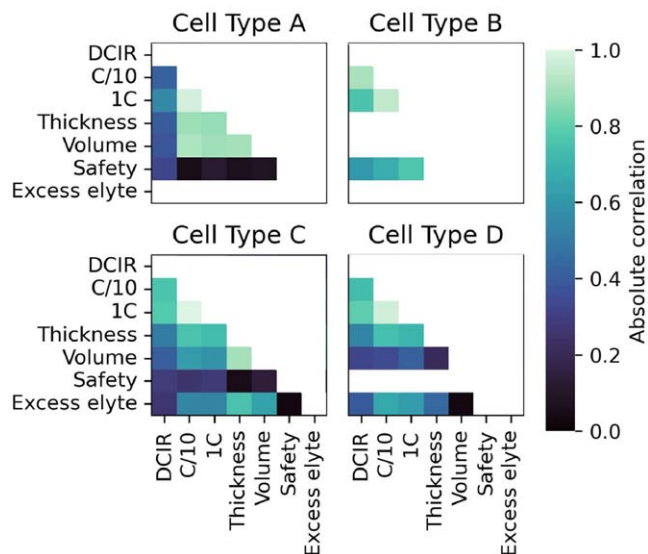


Figure 4. Absolute correlations between DCIR at 50% SOC and battery health and safety target variables. “DCIR” is resistance calculated using Ohm’s law on HPPC pulses conducted at the 2 C rate. “C/10” and “1 C” correspond to the constant-current discharge capacity at the noted rates. “Thickness” is the relative electrode stack thickness growth. “Volume” is the relative cell volume growth. “Safety” is the safe/unsafe classification quantified from post-charge voltage relaxation described in section 2.1.2. “Excess elyte” is the yes/minimal/no classification of excess electrolyte presence from XCT measurements. Supplemental Fig. S7 plots the same heatmaps with annotated values. For cell type B, high correlated of “safety” with DCIR and capacity is expected to be due to autocorrelation. Notice that the colormap chosen naturally lends itself to 3 easily distinguishable shades: light greens for strong correlations, blue/purple for weak correlation, and black for no correlation.

scores. MAPE approaching 0 and an R^2 approaching 1 correspond to a well performing model. MAPE is reported rather than mean absolute error (MAE), as each target has different units and the magnitude of the target variable may be different for each cell type, so comparisons across targets and cell types can only be made using unitless metrics.

FCR cycle charge efficiency and SOC are the easiest targets to predict. FCR cycle charge efficiency is likely easy to predict as the FCR cycle is qualitatively very similar to the input DC pulse sequence data, that is, the FCR cycle is basically 3 hours of DC pulses. Parity plots of the FCR cycle charge efficiency show quite good model performance for cell types A and C across all train/test splits and the entire range of measured values (Figs. 6L, 6N). SOC is easy to predict in comparison to discharge capacity, as SOC can be fully determined without any estimate of how much energy is actually in the battery. For example the OCV of the battery at rest is often used to estimate SOC, but this requires zero knowledge of the size of the battery. At minimum, two independent SOC measurements are required to estimate capacity.

In contrast, C/3 discharge capacity and US06 drive cycle charge throughput are relatively challenging to predict. Discharge capacity prediction is best for the C and D type cells, with C type cells having the lowest MAPE (2%–6%, average 4%) and D type cells having the highest R^2 (80%–95%). This accuracy is within the range of industrial battery management systems⁴³ and, while not perfect, demonstrates that discharge capacity prediction from rapid electrochemical measurements is accurate enough to be useful. However, the parity plots for cell types A and C (Figs. 6D, 6F) make it clear that while average error might be low, maximum error may be unacceptably high at the extremes; this error may be reduced with a more uniformly distributed data set, as for cell type D (Fig. 6G). The models are no better at predicting US06 charge throughput than they are the C/3 discharge capacity. Interestingly, it appears that capacity prediction for LFP-Gr cell is actually easier than SOC prediction; for only cell type D, MAPE is substantially lower for the discharge capacity prediction than for the SOC prediction.

The performance of tuned XGBoost models for predicting safety and physical targets is reported in Table II. As suggested by the data exploration (Fig. 4), these targets are not well correlated to DCIR, and the models trained on the raw I-V-T-t data do not make accurate predictions on test data. Prediction of the “safety” classification, which is perhaps the most critical metric to predict in addition to discharge capacity, completely fails, with an F1 score of zero for predicting unsafe cells for both cell types A and C. F1 score is the harmonic of precision (proportion of true predictions that are actually true) and recall (proportion of true values that are correctly predicted), so this score means that these models will essentially

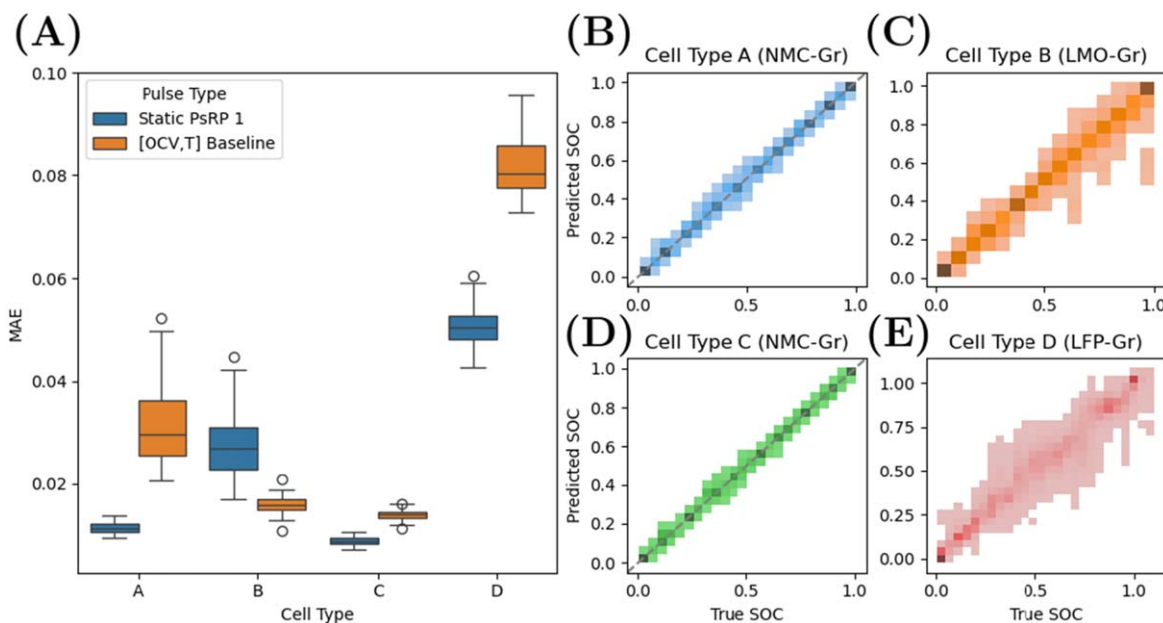


Figure 5. SOC prediction results. (A) Distribution of scores for each cell type, comparing the Static PsRP 1 pulse and a simple [OCV, T] as input features. (B-E) 2D binned histograms of true vs predicted values for each cell type using Static PsRP 1 pulses as input. Darker shades reflect a higher density of points.

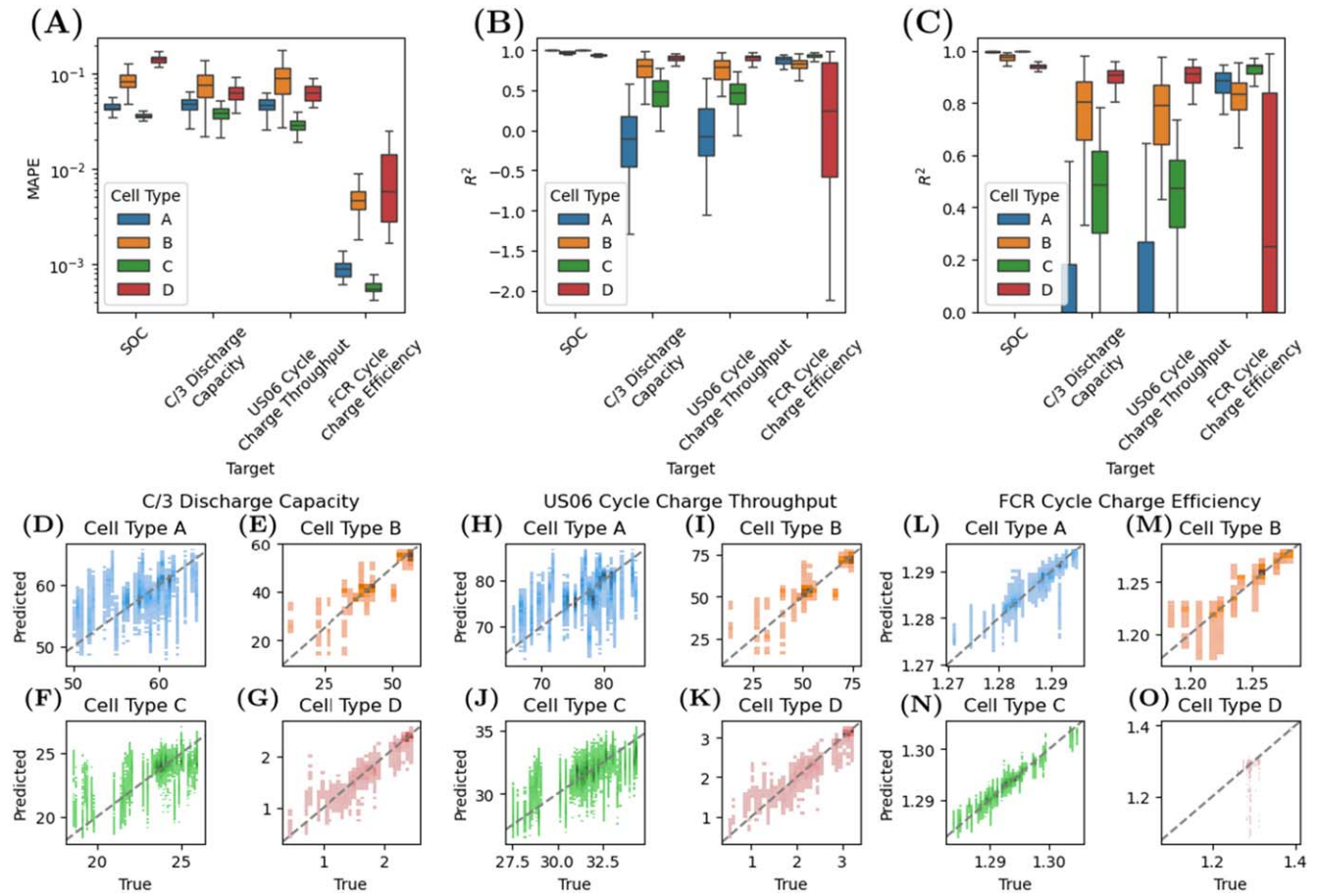


Figure 6. Comparing performance when predicting various targets, for all cell types, using Static PsRP 1 pulses. (A) Distribution of mean absolute percentage error (MAPE) when predicting different targets. (B) R^2 scores when predicting different targets for each cell type, with (C) clipped y-axis of R^2 scores for visibility of trends. (D-G) True and predicted C/3 discharge capacity (H-K) True and predicted US06 cycle charge throughput (L-O) True and predicted FCR cycle charge efficiency for each cell type using the Static PsRP 1 pulse input feature.

Table II. Performance when predicting safety and physical targets including volume growth, thickness growth, and presence of excess electrolyte using Static PsRP 1 pulses. F1 and R^2 scores > 0.5 are considered useful. All type D cells were considered “safe” so no model was trained to predict the “Safety” target for type D cells.

Target	Safety	Volume growth	Thickness growth	Excess electrolyte
Relevant metric	F1 score	R^2 score	R^2 score	F1 score
Cell Type A	0 (Unsafe) 0.89 (Safe)	−0.13	−0.31	N/A
Cell Type B	0.47 (Unsafe) 0.90 (Safe)	N/A	N/A	N/A
Cell Type C	0.01 (Unsafe) 0.90 (Safe)	−0.48	−0.46	0.45 (Excess) 0.44 (Minimal Excess) 0.09 (No Excess)
Cell Type D	N/A	−0.40	−0.11	0.77 (Excess) 0.77 (No Excess)

never predict a cell as “unsafe” correctly. This result is uncertain due to the unbalanced data set; with more “unsafe” samples, the data set would be more balanced, and the results more certain. However, volume and thickness growth predictions are similarly poor, with negative R^2 scores, despite the relatively uniform distribution of volumes and thicknesses and the relatively good correlation between thickness growth and capacity. Of all the safety and physical targets tested, only the presence or absence of excess electrolyte for cell type D can be predicted with anything even approaching a useful

accuracy. We are unsure of the mechanism by which excess electrolyte influences electrochemical response during DC pulses; the influence of excess electrolyte on electrochemical behavior has been of much recent study^{37,44,45} but it is unclear how those findings apply to cylindrical LFP-Gr Type D cells. These results make it clear that in addition to rapid electrochemical characterization, other electrochemical or physical characterization methods need to be used to validate the reliable and safe operation of battery systems prior to repurposing.

Complex pulses contain more information than simple pulses.—Figure 7 compares the predictive performance of XGBoost models using various pulse types on several different targets to investigate the impact of DC pulse sequence design on predictive performance. Models using the raw I-V-T-t data from various pulses are compared to two baselines: the previous [OCV, T] baseline, as well as a baseline that uses the OCV, initial temperature, and DCIR calculated using Ohm's law from the HPPC pulses as the sole features to illustrate the benefit of simply training on the raw I-V-T-t data. Overall, there is a small but consistent improvement to all predictions when moving from the “HPPC” type pulse to the slightly more complex “PsRP” pulse. These trends are most clearly seen for models that predict SOC for cell type D. Comparison of the “HPPC” pulse model performance to the DCIR baseline model performance across all plots also shows a consistent improvement, despite both models using identical raw data, demonstrating the straightforward effectiveness of the “just give the model all of the data” approach when there is enough data, rather than extracting features from the data using domain-specific models. Dynamic PsRP pulses have the lowest average error, as in addition to probing cell dynamics these pulses also charge or discharge the cell, which leads to measurable changes in cell OCV during the pulse sequence (3% nominal SOC change at the 1 C rate) and gives more information about overall cell state than a Static pulse, which has no net SOC change.

Systematic errors due to health, temperature, and pulse current.—Systematic error is model error that is not random, that is, when error is correlated with some conflating factor, like larger

error when measurements are at low SOC or high temperatures. Practical implementation of any predictive model requires understanding systematic errors that the model may suffer from to foresee potential problems. Figure 8 shows the systematic errors of models using the Static PsRP 1 pulse sequence as input with respect to the predicted target variable. Systematic error plots of the discharge capacity prediction (Figs. 8A–8D) show a somewhat concerning overall trend of increasing error at decreasing capacity. This issue can likely be mitigated by collecting more training data from low capacity cells to create a more uniformly distributed training data set, as shown by the low systematic error of cell type D (Fig. 8D). Systematic errors of FCR cycle charge efficiencies behave similarly, with low values from more degraded cells predicted with higher average error. SOC prediction (Figs. 8I–8L) instead shows higher average error in the middle of the SOC range, where the voltage curves for all cell types are relatively flat compared to transport-limited regions at the extreme ends of the SOC curve.

Figure 9 shows the systematic errors of the 1 C discharge capacity prediction with respect to exogenous variables, that is, variables not related to the internal state of the cell. Figure 9A shows that the model has learned how to compensate for the impact of temperature, with no preference for under- or over-prediction and high or low temperatures. Similarly, the impact of the SOC at which a pulse is conducted is entirely compensated for (Fig. 9D). By using HPPC pulses as input, the impact of pulse current can be examined (Fig. 9B,C). Increasing pulse current leads to a slight decrease of average error. This decrease of error is probably due to increasing signal-to-noise ratio, as the change of voltage after 10 s at the C/10 current may be only a bit larger than the 1 mV voltage resolution of

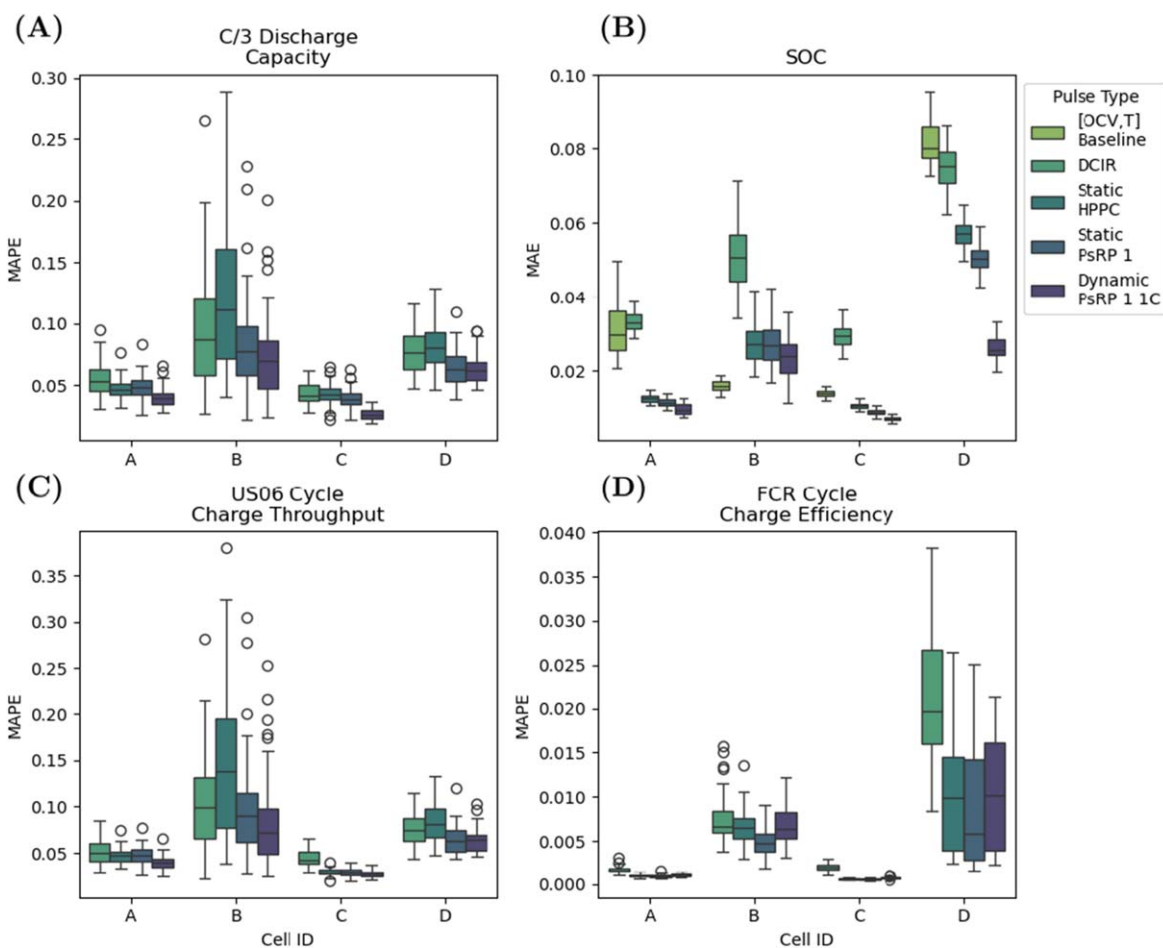


Figure 7. Comparing performance of different pulse types. (A) Distributions of MAPE when predicting C/3 discharge capacity. (B) Distributions of MAPE when predicting SOC, with an additional comparison to a “[OCV,T]” baseline model just using the initial voltage and temperature to predict SOC. (C) Distributions of MAPE when predicting US06 cycle charge throughput. (D) Distributions of MAPE when predicting FCR cycle charge efficiency.

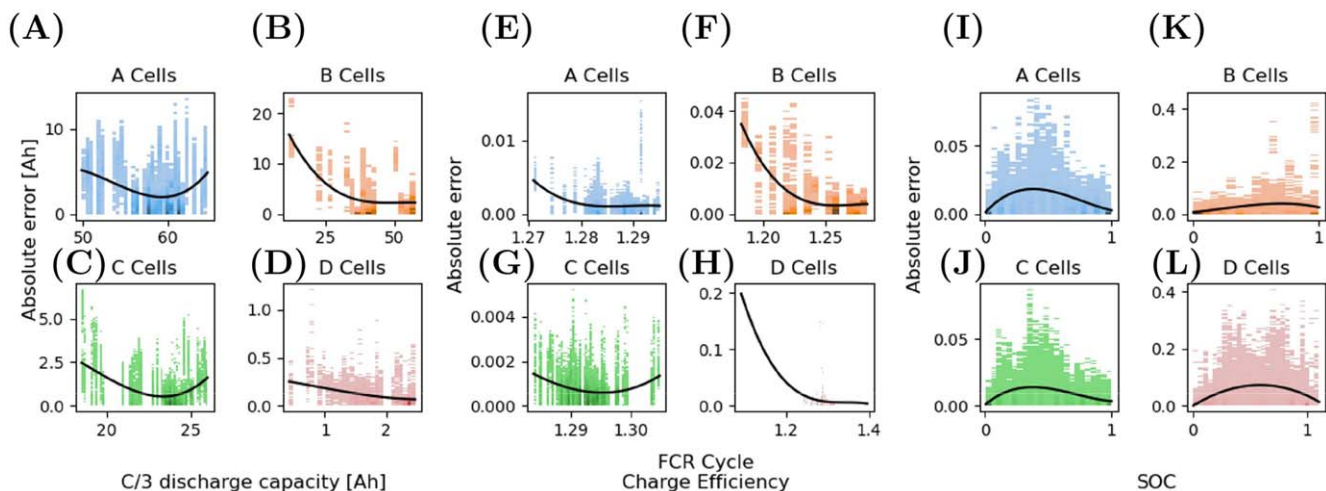


Figure 8. Systematic errors with respect to target values using Static PsRP 1 pulse as input. (A-D) Effect of C/3 discharge capacity on absolute error [Ah]. (E-H) Effect of FCR cycle charge efficiency on absolute error. (I-L) Effect of SOC on absolute error.

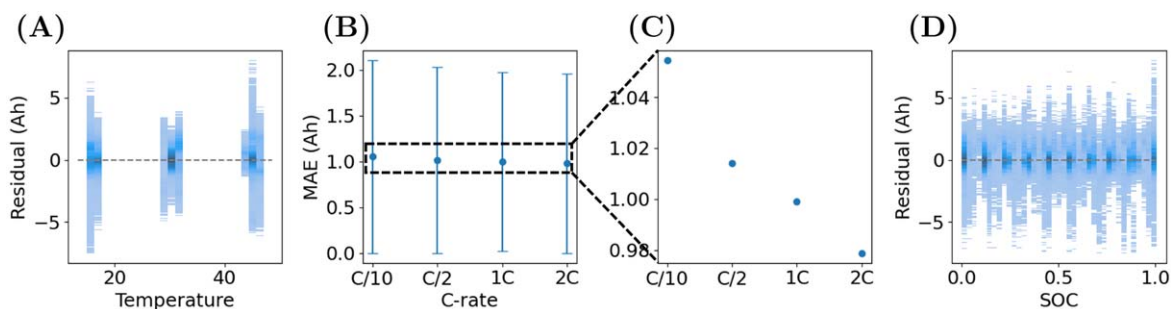


Figure 9. Systematic error of 1 C discharge capacity prediction with respect to exogenous variables. Plots show errors of 1 C discharge capacity predictions using Static HPPC pulses on cell type C. (A) Effect of temperature on residual error. (B) Effect of C-rate on MAE, (C) zoom view without error bars to highlight the trend. (D) Effect of SOC on residual error.

the testing channels. Even higher currents than those studied here may drive new physics, which may increase or decrease accuracy, but also increases the risks of catastrophic failures.

Discussion

Usefulness of capacity prediction as a rapid screening method for second-life sorting.—To demonstrate the practical usefulness of capacity prediction using only a 120 s DC pulse sequence, we treat capacity prediction as a classification problem, predicting capacity as “good”, “okay”, or “bad” to simulate how cells may be screened for disposal or reuse using rapid electrochemical characterization tests. Note that we are not suggesting to sort cells based on a rapid electrochemical measurement alone, but rather as a screening step to prevent more time-consuming characterization methods from being wasted on poor performing cells. Results from the capacity classification problem are shown in Fig. 10. Results are extremely promising: given training data from cells with relatively well-distributed capacities, good and bad cells can be readily separated with near-zero errors. In the case of the type D cells (10E), there are actually zero cases of confusion between good and bad cells across all 50 random train/test splits, demonstrating the impact of having a high quality data set to train from. Considering a real-world cell sorting plant, the critical failure to avoid are false-positives, i.e., cells that are predicted “good” but are actually “bad”. Across all cell types, the “good” to “bad” false positive rate is just 0.3%, and the classification accuracy is 99.5%. Expanding to separation of “good” and “okay” from “bad” cells, the false positive rate is an order of magnitude higher at 2.7%, and the classification accuracy is down to 95%. The huge majority of classification errors here are between

“okay” and “good” cells: cell resistance often does not change dramatically throughout much of a battery’s operating life,^{9,22} making the capacity prediction problem at capacities greater than 80% extremely challenging.

While we demonstrate here that cells can be sorted by capacity using DC pulses, it is not just sufficient to measure capacity to certify cells for reuse in a new application; safety and reliability of cells also has to be diagnosed. This is discussed in a following section.

AC pulses do not appear to be any more useful than DC pulses.—While DC pulses and AC electrochemical impedance spectroscopy (EIS) are fundamentally probing the same physical systems, they may differ in their sensitivity to cell behaviors in different frequency/time-constant regimes,⁴⁶ and each method has its own various pros and cons for practical implementation. Only DC pulses were used here because the battery cyclers used could not perform EIS, so we cannot make a direct comparison. However, there are several open-source data sets with EIS-capacity feature-target pairs available, and we can compare the predictive performance from these works to our own. The four data sets used here are reported in Table III. For data sets collected by measuring EIS multiple times on each cell (Zhang et al.,¹⁰ Fajiri-Niri et al.,¹¹ Gasper et al.⁹), prediction error is reported from leave-one-out cross-validation with measurements grouped by cell. For the Ramirez-Meyers et al. data,¹² where there is only one EIS measurement for each cell, 5-fold cross-validation was performed. Every data set is modeled using an XGBoost model using a feature selection step that uses the impedance from only the two best frequencies, based on experience from prior work that demonstrated that this feature selection strategy reliably resulted in the best test set prediction

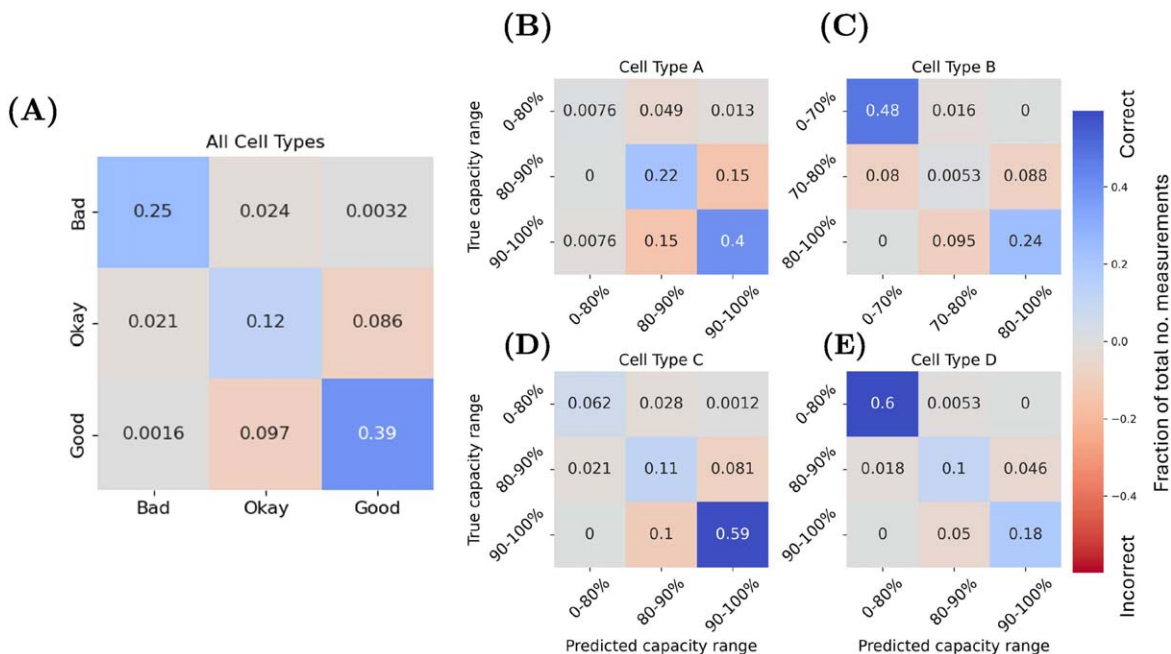


Figure 10. Confusion matrix of C/3 discharge capacity classification models. All models use Static PsRP 1 pulses as input. (A) Summary of C/3 discharge capacity classification from all cells. For cell types A, C, and D, “Good” is greater than 90%, “Okay” is less than 90% but greater than 80%, and “Bad” is less than 80%. For B type cells, thresholds between “Good”, “Okay”, and “Bad” are at 10% lower capacity (80% and 70%, respectively) to create a more balanced classification problem for that cell type. (B-D) Confusion matrices for the C/3 discharge capacity classification problem for cell types A, B, C, and D separately.

Table III. Performance of relative capacity prediction models using EIS data from various data sets. All errors are reported for test predictions made during cross-validation.

Data set	# Measurements	MAPE	R ²
Zhang et al. ¹⁰	1343	0.082	0.80
Faraji-Niri et al. ¹¹	360	0.019	0.89
Gasper et al. ⁹	467	0.032	0.74
Ramirez-Meyers et al. ¹²	198	0.092	0.83

performance.⁹ The two selected frequencies as well as a few XGBoost hyperparameters were optimized through the same cross-validation scheme as above for every training fold.

Results from this analysis are shown in Fig. 11 and Table III. The parity plots as well as the predictive metrics are similar to the results reported for type C and D cells here, suggesting that the two rapid characterization methods (DC pulse sequences and AC EIS) are equally as useful for predicting capacity. In particular, note that the type D cells studied in this work are a subset of the cells reported in Ramirez-Meyers et al.,¹² and the EIS modeling result lies well within the distributions shown in Fig. 6.

We do not have the data to comment on whether EIS may be more useful for predicting safety or reliability targets than DC pulses.

Rationalizing why state-of-charge and available discharge capacity are easier to predict than total discharge capacity.—As shown in Fig. 6 and discussed previously, SOC prediction is approximately 2–3 x “easier” than the prediction of discharge capacity. Given this, a relatively obvious approach to diagnosing discharge capacity would simply be to predict SOC again after some capacity has been charged or discharged, then use the predicted Δ SOC and measured charge throughput to estimate overall discharge capacity. However, because SOC needs to be estimated twice, the error of these estimates is multiplied. This can be demonstrated using straightforward calculations, with results shown

in Fig. 12. Given the 1% average SOC estimation error we actually achieved for type C cells (Fig. 5), this example shows we would need a Δ SOC swing of about 50% to get an SOH estimate with less than 4% error (similar to the error of capacity prediction for type C cells, see Fig. 6). To get a useful SOH estimate with a smaller Δ SOC swing, say 20%, we would need an extraordinarily accurate SOC estimate (0.1% average error).

Basically, this thought experiment is simply to show that SOC prediction may “look” better than capacity prediction, but this is simply a result of the prediction problem being fundamentally easier. Similarly, it is straightforward to infer that prediction of “available discharge capacity” or “remaining discharge energy”, that is, the amount of energy that could be discharged at some rate from the cell’s current state, is also easier than predicting overall capacity; available discharge capacity can be predicted without knowing current SOC, and thus without knowing the overall capacity of the cell. Recent works have demonstrated about 3% average error for remaining discharge energy prediction,⁴⁷ which lies nicely between the 1% average error for SOC prediction and 5% average error for capacity prediction shown in this work (cell type C).

Challenges for quantifying and diagnosing battery safety.—As shown in Table II, safety and reliability relevant cell properties cannot be accurately predicted using rapid DC pulse sequences, necessitating other characterization techniques to validate the safety and reliability of a battery before reuse. Of course, safety diagnosis could be performed with the same methods used here, for example, fully charging the battery system at a 1 C rate at low temperature and using the post-charge voltage relaxation data to evaluate safety. However, this approach risks catastrophic failure, especially when conducted on battery systems rather than individual cells. So, in addition to further electrochemical characterization after rapid screening, like conducting a full charge/discharge cycle, other diagnostic methods are required to ensure safe and reliable future operation.

Techniques for diagnosing cell safety for second-life applications should meet a few key requirements.⁴⁸ First, the technique should be nondestructive both on the cell level and pack level. Second, the technique should be able to inspect the entire cell and/or system as

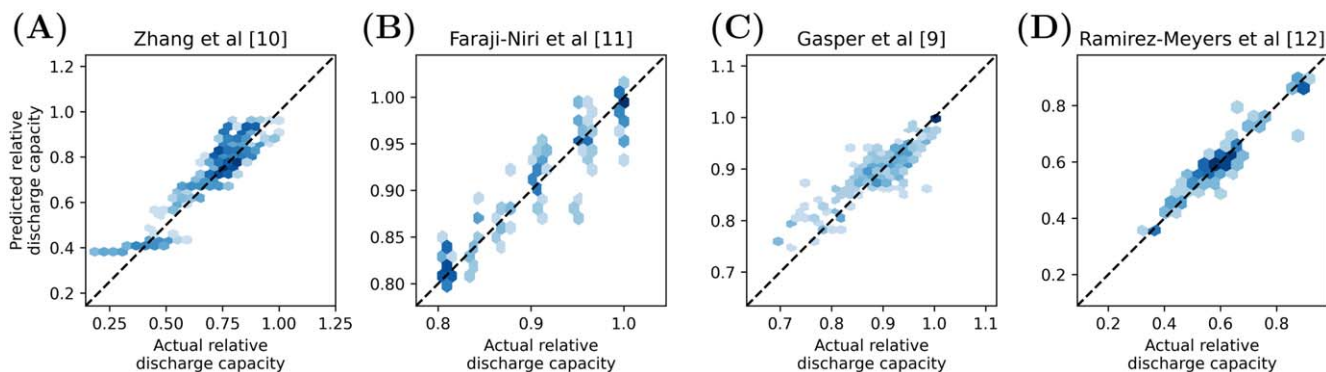


Figure 11. Prediction of relative discharge capacity from EIS on four public data sets. (A) Zhang et al.;¹⁰ EIS at 100% SOC and [25, 35, 45] °C. (B) Faraji-Niri et al.;¹¹ EIS at [5, 20, 50, 70, 95]% SOC and [15, 25, 35] °C. (C) Gasper et al.;⁹ EIS varying temperatures and SOC, but most data at [50, 90]% SOC and [−10, 0, 25] °C. (D) Ramirez-Meyers et al.;¹² EIS after discharge and at room temperature.

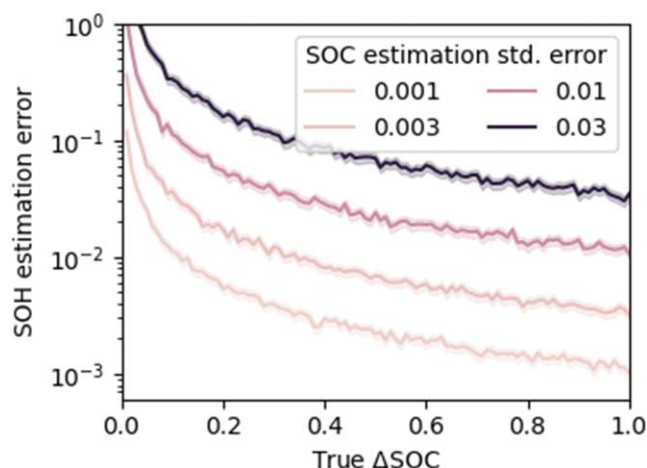


Figure 12. Simulating the error of estimating SOH from two SOC estimates. Average error (line) and 95% confidence interval (shaded region) from 200 random samples assuming a normally distributed error for both upper and lower SOC estimates at each Δ SOC.

defects can occur anywhere, from within individual electrode layers in a cell to the inter-cell connections or other system-level defects. Third, the technique should have high spatial resolution (ideally ≤ 50 μm) to detect latent defects that can induce safety issues. Lastly, the technique must be rapid (ideally ≤ 10 seconds per cell) to meet the challenging throughput requirements for this application. Unfortunately, few methods can meet all of these requirements today—especially the requirement for avoiding pack disassembly. Simple techniques such as “gas sniffing” (detecting electrolyte or electrolyte degradation product molecules from the gasses in or around the battery system) can be readily applied but provide limited diagnostic utility. Other methods like magnetic resonance and acoustics are rapid and can scale to large systems but have limited spatial resolution. In our view (although please see our Declaration of Interests statement), XCT reveals a wealth of insights into battery quality due to its high spatial resolution and has the potential for high scalability.^{48–50} However, disassembling packs down to the cell level as required for XCT is not feasible for most pack architectures. Ultimately, a combination of inspection techniques on a sampling basis (in-depth characterization may be infeasible on all samples) may be required for comprehensive defect and safety screening for second-life applications.

Challenges for pack state estimation.—Pack performance is due to a combination of both the average cell performance as well as the variance of cell performance, i.e., pack imbalance. Here, we only

demonstrate predicting cell performance, but these methods may extend well to also predicting pack imbalance. When considering how these methods may work at pack scale, there are both advantages and disadvantages when moving from cells in the lab to packs in the real world.

The main disadvantages when scaling to the pack level are the absence of current measurements for individual cells in parallel groups and the loss of sensitivity due to cell-to-cell variability. As to the first disadvantage, we tested simply training models without any current data and showed only a small loss of accuracy in the resulting predictions (Supplemental, Table 1). However, simply removing current as an input feature neglects the fact that not only is the magnitude of the current between parallel cells unknown, but current is also dynamic and variable across cells in the parallel group⁵¹ (a constant current across the parallel group will necessarily be constant current for each individual cell), which we did not emulate when generating this data set. This variability will “blur out” the voltage response of the battery pack, likely decreasing prediction accuracy.

As to the advantages, battery packs come with data related to historical use as well as active estimates of state-of-charge and state-of-health, all tracked and reported by the BMS.⁵² For instance, the California “Data Standardization Requirements for 2026 and Subsequent Model Year Light-Duty Zero Emission Vehicles and Plug-in Hybrid Electric Vehicles”⁶ requires BMS reporting of health within $\pm 5\%$ of actual measurable capacity and historical data like total distance traveled, total active/idle time, average temperature, and time at low/mid/high SOC, all of which can help to improve accuracy of health and remaining useful life estimation if detail about the degradation behavior of that specific battery system is known. Also, while cell-to-cell variability may decrease prediction accuracy, the variability of voltage response from each cell or parallel group in series gives more information about pack health, and may be used to quantify cell-to-cell variability or diagnose module- and system-level performance in addition to cell-level performance.^{17,19}

Another key challenge for scaling from cell to pack level is acquiring training data. Here, tens of individual cells with different states of health and use histories were used to train machine learning models; acquiring this same data at the pack level is dramatically more expensive. As the repurposing industry matures, and more information about the aging behavior of specific battery systems becomes available publicly or privately, this may not be a challenge. But at the moment, these data limitations are a barrier to the use of these methods by anyone but the original battery system manufacturers and operators. It may be possible to use cell-level measurements to inform system-level models⁵³ to generate synthetic data for training machine-learning models, but great care would need to be taken to validate the accuracy of the synthetic data before trusting this approach.

Concluding Remarks

We can summarize the results of our investigation on the usefulness of rapid electrochemical diagnostics methods into straightforward positive and negative findings. On the positive side, rapid diagnostics enable accurate SOC and capacity prediction, with enough fidelity to enable capacity screening by a battery repurposing facility or to complement on-line SOC and SOH estimation in battery systems. On the negative side, capacity estimation is probably not accurate enough to enable remaining useful life estimation, and we show a complete failure to detect any potential safety and reliability issues that we identified via post-charge voltage relaxation analysis, physical inspection, and XCT scanning. These findings highlight the critical need for the development of fast and accurate methods for non-destructive safety and reliability diagnosis of battery systems.

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Author Contributions

Conceptualization: P.G.; Data curation: P.G., N.P., B.K.; Formal analysis: P.G., N.P., B.K., P.M.A.; Funding acquisition: P.G., M.K.; Investigation: P.G., B.K., T.B., A.C.; Methodology: P.G., B.K., N.P.; Project administration: P.G., M.K.; Software: P.G., N.P.; Resources: M.K., K.R.-M., P.M.A.; Supervision: P.G., M.K.; Visualization: P.G., B.K., N.P.; Writing - original draft: P.G., B.K., N.P.; Writing - review & editing: P.G., B.K., N.P., P.M.A., K.R.-M.

Declaration of Interests

A.C. and P.M.A. have a financial interest in Glimpse Engineering Inc., which specializes in high-throughput CT scanning for batteries.

Materials Availability

Unless we observed safety issues following testing, such as excessive gas generation, all tested batteries have been kept fully discharged in a freezer since the completion of this study. Type B, C, and D batteries reported in this study may be available to other researchers upon request.

Data and Code Availability

Code for data processing, modeling training, and analysis is available on Github (https://github.com/NREL/battery_pulse_diagnostics). Electrochemical data is too large to be hosted on Github, but is available on Zenodo along with the code (<https://doi.org/10.5281/zenodo.14597394>).

XCT scans from type C and D cells can be viewed using the Glimpse Portal (<https://app.glimp.se>).

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